

Malware Network Analysis

Project Overview: I utilized Wireshark to analyze real-time malware network traffic.

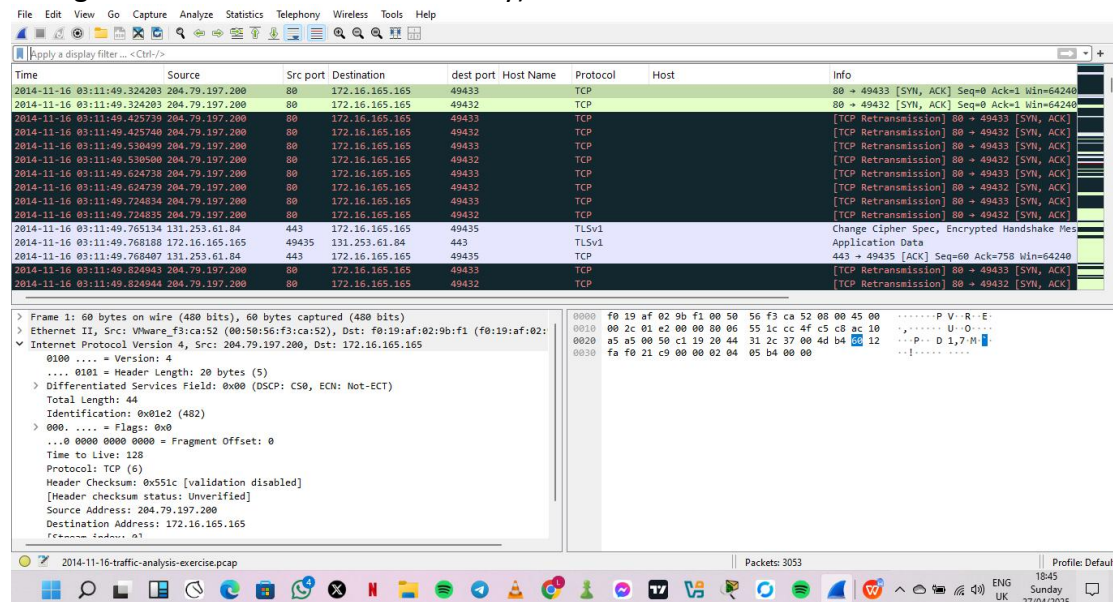
Objectives: To identify the following Indicator of Compromise;

1. What are the infected file(s) downloaded and their hashes?
2. What is the URL / domain of the infected site?
3. What is the IP address of the infected website?
4. What is the IP address of the infected machine?
5. What is the hostname of the infected machine?
6. What is the mac address of the infected machine?

Tools Used: Wireshark, Malware file, Hashmyfiles, Virustotal.com
(Link to malware file will be in my github repository)

Instructions:

Start Wireshark > open the malware file > configure your wireshark for easier investigation just like the screenshot below (remove number of packets, length, change time to Date & Time of the Day).



Click on Statistics, select protocol hierarchy to view where the most traffic comes in and help hasten our investigation process like in the screenshot below it is Internet Protocol 4 the in it there is Transport Control Layer(TCP) which has Hyper Text Transfer Protocol(HTTP) > Apply as filter(either by right clicking it and select apply as filter;selected or close the the tab and enter http in the filter field) > then enter http.request in the filter field, this will only show HTTP GET & POST request only.

The image displays the Wireshark network protocol analyzer interface. The top pane shows the 'Protocol Hierarchy' for a capture file named '2014-11-16-traffic-analysis-exercise.pcap'. The hierarchy is expanded to show the 'Internet Protocol Version 4' (IPv4) section, which includes 'User Datagram Protocol' (UDP) and 'Transmission Control Protocol' (TCP). The 'TCP' section is further expanded to show 'Hypertext Transfer Protocol' (HTTP) and 'Media Type'. The bottom pane shows the 'Packet Details' for a selected packet (Frame 52), which is an 'HTTP Request' to 'www.bing.com'. The details pane shows the 'Request Line' as 'POST /fd/ls/lsp.aspx HTTP/1.1' and the 'Host' as 'www.bing.com'. The 'Packet Bytes' pane shows the raw packet data in hexadecimal and ASCII.

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU/s
Frame	100.0	3053	100.0	2502525	30 k	0	0	0	3053
Ethernet	100.0	3053	1.8	44068	537	0	0	0	3053
Internet Protocol Version 6	0.7	22	0.0	920	11	0	0	0	22
User Datagram Protocol	0.6	17	0.0	136	1	0	0	0	17
Link-local Multicast Name Resolution	0.2	6	0.0	164	1	6	164	1	6
DHCPv6	0.4	11	0.0	1012	12	11	1012	12	11
Internet Control Message Protocol v6	0.2	5	0.0	140	1	5	140	1	5
Internet Protocol Version 4	98.7	3012	2.4	60260	734	0	0	0	3012
User Datagram Protocol	1.8	56	0.0	448	5	0	0	0	56
NAT/OS Name Service	0.8	24	0.0	1524	18	24	1524	18	24
Link-local Multicast Name Resolution	0.2	6	0.0	164	1	6	164	1	6
Dynamic Host Configuration Protocol	0.1	4	0.0	1208	14	4	1208	14	4
Domain Name System	0.7	22	0.0	1069	13	22	1069	13	22
Transmission Control Protocol	96.7	2951	2.4	60540	738	2501	51540	628	2951
Transport Layer Security	12.2	372	19.5	487155	5939	372	469245	5720	387
Hypertext Transfer Protocol	2.6	78	51.5	1289552	15 k	41	17935	218	78
Media Type	0.3	8	50.4	1260641	15 k	8	1260641	15 k	8
Line-based text data	0.5	15	28.7	717681	8749	15	717681	8749	15
JPEG File Interchange Format	0.1	2	1.2	30362	370	2	30362	370	2
eXtensible Markup Language	0.1	3	0.1	2092	25	3	2092	25	3
CompuServe GIF	0.3	9	3.2	80607	982	9	80607	982	9
Internet Group Management Protocol	0.2	5	0.0	80	0	5	80	0	5
Address Resolution Protocol	0.6	19	0.0	532	6	19	532	6	19

Add more column to the interface by right clicking and select column preference then add source port (it shows where the message is coming from) and destination port (where the message needs to go) just like the screenshot below.

Wireshark packet capture showing HTTP requests. The packet list displays several GET requests to various domains, including www.bing.com and www.ciniholland.nl. The packet details pane shows the structure of an HTTP request, including the header and body. The packet bytes pane shows the raw data in hexadecimal and ASCII.

To find the infected files click file;export object;http to view the files downloaded by the infected computer, from my findings only the files that looks malicious which is the java file, the .exe file and .swf file since they need to be executed by victim, save the three files in a folder(do not execute on your computer)

Wireshark packet capture showing HTTP requests. The packet list displays several GET requests to various domains, including www.ciniholland.nl and other domains. The packet details pane shows the structure of an HTTP request, including the header and body. The packet bytes pane shows the raw data in hexadecimal and ASCII.

Now, that since the malwares has been saved, proceed to find the indicator of compromise in the objectives above;

1. To find the hash values, download hashmyfiles from their website (https://www.nirsoft.net/utils/hash_my_files.html) open the app and add the three files to generate their hash values, visit virustotal.com, input and search their hash value to see if the files is infected or not. Screenshot below

HashMyFiles						
File Edit View Options Help						
File Edit View Options Help						
Filename	MD5	SHA1	CRC32	SHA-256	SHA-512	SHA-384
index.php%3fref=jar...	1e34fdebf655cebea78b45e43520ddf	8bc0077afbcf1f19cdc7a3fec0d145bfb9975d0	0116a18e	178be0ed83a7a9020121dee1c305fd6ca3b74d15836835cfb1684da0b44190d3	8709d1751cd21da812a39066a3d09d3716c2d...	0ba8eed737b73e3a...
index.php%3fref=mp...	d276c86dcdbcd6b74ee02496bc90d98	640c8a2ecde8af90b987a51f2fa59c7c8dfa8fc	94af21ee	ce5d1c41f6ca739f6c69f175a8f688334df2ea4...	4a48a940fc360d92facfd26a033a2117b1c90f...	a8114dd28d079a5f...
index.php%3fref=swf...	7b3baa7d6bb3720f369219789e38d6ab	630f030ca896412cb460dea77973c67fa316c2ce	cacde55a	e2e3b802a0d939d07bd8291f23484c2f68ccc...	cb1dbde708bf44dfdfa7d9231b98358be0222...	fc3aaa39e5c1afbf1...

3 file(s)

NirSoft Freeware. <https://www.nirsoft.net>

178be0ed83a7a9020121dee1c305fd6ca3b74d15836835cfb1684da0b44190d3

34 / 63

Community Score

-34

34/63 security vendors flagged this file as malicious

Reanalyze Similar More

178be0ed83a7a9020121dee1c305fd6ca3b74d15836835cfb1684da0b44190d3

Size 10.36 KB

Last Analysis Date 1 day ago

JAR

jar exploit cve-2012-0507 checks-cpu-name detect-debug-environment sets-process-name

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 6

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label trojan:java/tafold Threat categories trojan Family labels java rtfold expjava

Security vendors' analysis Do you want to automate checks?

Alibaba	Exploit:JAVA/CVE-2012-0507.ea078876	AliCloud	Exploit:Java/Agent_RAW
ALYac	Java.Trojan.GenericGB.29292	Arcabit	Trojan.Generic.D1ECC17A
Avast	Java/Malware-gen (Trj)	AVG	Java/Malware-gen (Trj)

ce5d1c41f6ca739f6c69f175a8f688334df2ea4d2e0fb5e3243ae01efd251e81

0 / 61

Community Score

No security vendors flagged this file as malicious

Reanalyze Similar More

ce5d1c41f6ca739f6c69f175a8f688334df2ea4d2e0fb5e3243ae01efd251e81

Size 392.39 KB

Last Analysis Date 1 day ago

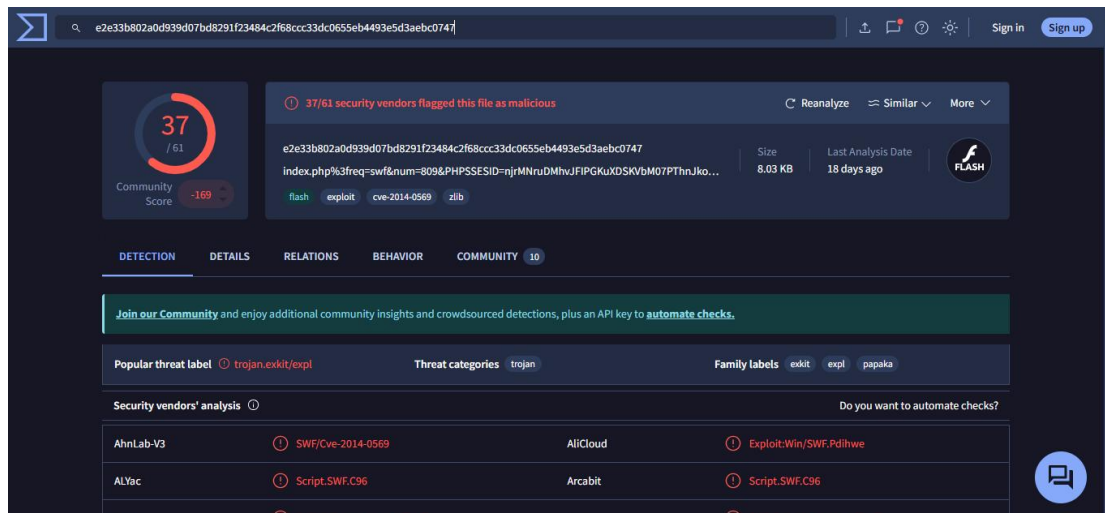
file5

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis Do you want to automate checks?

Acronis (Static ML)	Undetected	AhnLab-V3	Undetected
AliCloud	Undetected	ALYac	Undetected
Antiy-AVL	Undetected	Arcabit	Undetected
Avast	Undetected	AVG	Undetected



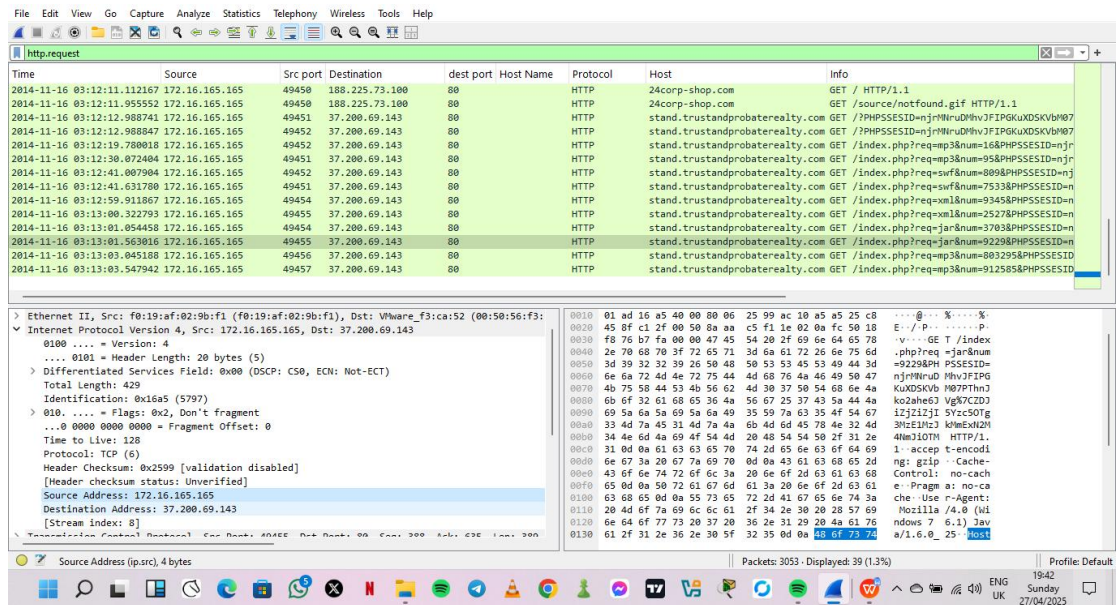
From the screenshot above only two files are tested infected the second one could be malware free or the virus isn't executed yet.

2. To find the URL/Domain of the Infected Site, trace it from the HTTP file in wireshark

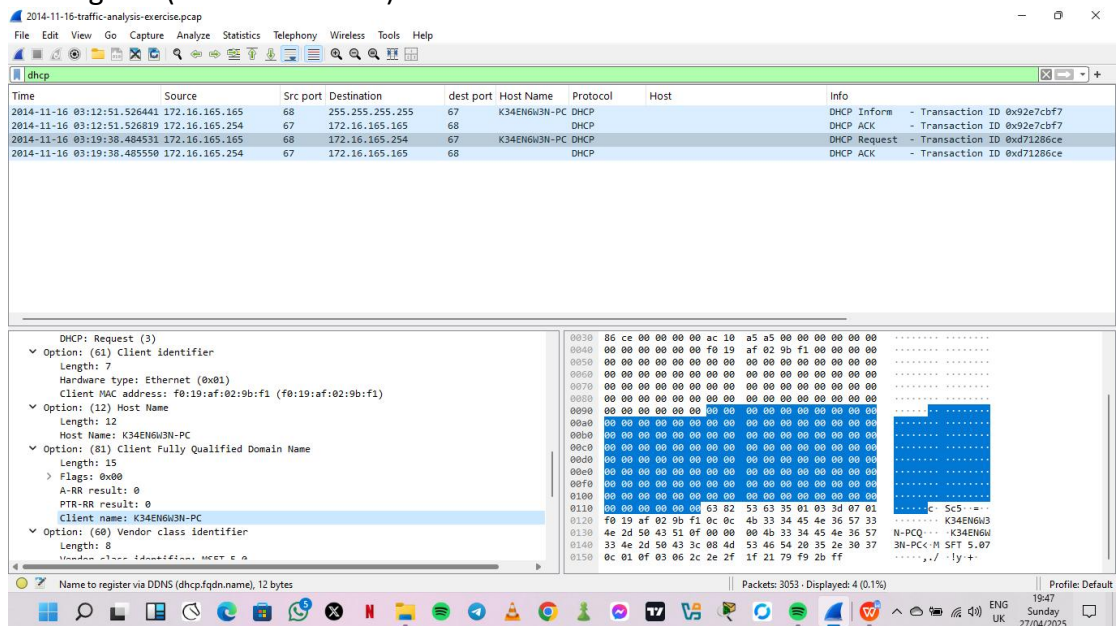
1554	stand.trustandprobaterality.com	text/html	257 KB	TPHPSSSEID=njrmNruDMhvJFIPGKuXDSKVbM07PThnko2ahe6VgZDIZJZjZj5Vzc50Tg3MzE1MzJkMmExN2M4NmIOTM
1566	stand.trustandprobaterality.com	text/html	255 KB	TPHPSSSEID=njrmNruDMhvJFIPGKuXDSKVbM07PThnko2ahe6VgZDIZJZjZj5Vzc50Tg3MzE1MzJkMmExN2M4NmIOTM
1991	stand.trustandprobaterality.com	application/x-msdownload	401 KB	index.php?req=mp3&num=16&PSSSEID=njrmNruDMhvJFIPGKuXDSKVbM07PThnko2ahe6VgZDIZJZjZj5Vzc50Tg3MzE1MzJkMmExN2M4NmIOTM
2379	stand.trustandprobaterality.com	application/x-msdownload	401 KB	index.php?req=mp3&num=95&PSSSEID=njrmNruDMhvJFIPGKuXDSKVbM07PThnko2ahe6VgZDIZJZjZj5Vzc50Tg3MzE1MzJkMmExN2M4NmIOTM
2394	stand.trustandprobaterality.com	application/x-shockwave-flash	8227 bytes	index.php?req=swf&num=809&PSSSEID=njrmNruDMhvJFIPGKuXDSKVbM07PThnko2ahe6VgZDIZJZjZj5Vzc50Tg3MzE1MzJkMmExN2M4NmIOTM
2415	stand.trustandprobaterality.com	application/x-shockwave-flash	8227 bytes	index.php?req=swf&num=733&PSSSEID=njrmNruDMhvJFIPGKuXDSKVbM07PThnko2ahe6VgZDIZJZjZj5Vzc50Tg3MzE1MzJkMmExN2M4NmIOTM
2469	stand.trustandprobaterality.com	text/xml	572 bytes	index.php?req=xml&num=934&PSSSEID=njrmNruDMhvJFIPGKuXDSKVbM07PThnko2ahe6VgZDIZJZjZj5Vzc50Tg3MzE1MzJkMmExN2M4NmIOTM
2475	stand.trustandprobaterality.com	text/xml	572 bytes	index.php?req=xml&num=2527&PSSSEID=njrmNruDMhvJFIPGKuXDSKVbM07PThnko2ahe6VgZDIZJZjZj5Vzc50Tg3MzE1MzJkMmExN2M4NmIOTM
2489	stand.trustandprobaterality.com	application/java-archive	10 KB	index.php?req=jar&num=3703&PSSSEID=njrmNruDMhvJFIPGKuXDSKVbM07PThnko2ahe6VgZDIZJZjZj5Vzc50Tg3MzE1MzJkMmExN2M4NmIOTM
2502	stand.trustandprobaterality.com	application/java-archive	10 KB	index.php?req=jar&num=9229&PSSSEID=njrmNruDMhvJFIPGKuXDSKVbM07PThnko2ahe6VgZDIZJZjZj5Vzc50Tg3MzE1MzJkMmExN2M4NmIOTM
2977	stand.trustandprobaterality.com	application/x-msdownload	401 KB	index.php?req=mp3&num=803295&PSSSEID=njrmNruDMhvJFIPGKuXDSKVbM07PThnko2ahe6VgZDIZJZjZj5Vzc50Tg3MzE1MzJkMmExN2M4NmIOTM

3. To find the IP address of the infected website, open wireshark and click Internet Protocol Version 4 in the details pane and look for destination address (37.200.69.143)

4. To find the IP address of the Infected Machine, still in the Internet Protocol Version 4 section look for source address(172.16.165.165)



5. To find the HostName of the infected machine, search DHCP in the filter field then click DHCP request in the details pane expand DHCP request scroll and look for host name given (K34EN6W3N-PC)



6. To find MAC Address of the infected machine, still in dhcp request details pane scroll and find mac address(f0:19:af:02:9b:f1)

2014-11-16-traffic-analysis-exercise.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dhc

Time	Source	Src port	Destination	dest port	Host Name	Protocol	Host	Info
2014-11-16 03:12:51.526441	172.16.165.165	68	255.255.255.255	67	K34ENGW3N-PC	DHCP		DHCP Inform - Transaction ID 0x92e7cbf7
2014-11-16 03:12:51.526819	172.16.165.254	67	172.16.165.165	68		DHCP		DHCP ACK - Transaction ID 0x92e7cbf7
2014-11-16 03:19:38.484531	172.16.165.165	68	172.16.165.254	67	K34ENGW3N-PC	DHCP		DHCP Request - Transaction ID 0xd71286ce
2014-11-16 03:19:38.485550	172.16.165.254	67	172.16.165.165	68		DHCP		DHCP ACK - Transaction ID 0xd71286ce

Client MAC address (dhcp.hw.mac_addr), 6 bytes

Packets: 3053 · Displayed: 4 (0.1%)

Profile: Default

19:49 Sunday 27/04/2025

2014-11-16-traffic-analysis-exercise.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dhc

Time	Source	Src port	Destination	dest port	Host Name	Protocol	Host	Info
2014-11-16 03:12:51.526441	172.16.165.165	68	255.255.255.255	67	K34ENGW3N-PC	DHCP		DHCP Inform - Transaction ID 0x92e7cbf7
2014-11-16 03:12:51.526819	172.16.165.254	67	172.16.165.165	68		DHCP		DHCP ACK - Transaction ID 0x92e7cbf7
2014-11-16 03:19:38.484531	172.16.165.165	68	172.16.165.254	67	K34ENGW3N-PC	DHCP		DHCP Request - Transaction ID 0xd71286ce
2014-11-16 03:19:38.485550	172.16.165.254	67	172.16.165.165	68		DHCP		DHCP ACK - Transaction ID 0xd71286ce

Client MAC address (dhcp.hw.mac_addr), 6 bytes

Packets: 3053 · Displayed: 4 (0.1%)

Profile: Default

19:49 Sunday 27/04/2025

Key Findings:

- Insecure Internet Protocol.
- HTTP traffic with suspicious URL.
- Files being downloaded; malware pull down files from malicious and random servers.

Conclusion:

This analysis already gives the incident responder the clues they need to respond to the malware attack on the infected machine.